

SACHIN KURUP

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PROFILE SUMMARY

Final-year Computer Science student with a strong foundation in AI research, seeking to pursue doctoral study in agentic AI systems and multi-agent architectures. My research interests centre on the design and evaluation of retrieval-augmented generation frameworks, decision-aware agents, and cross-lingual NLP; spanning both theoretical inquiry and empirical validation. I have contributed to interdisciplinary research environments at Amrita Vishwa Vidyapeetham and Colorado State University, and have authored and co-authored multiple publications across ACL workshops, IEEE symposia, and international conferences. Alongside my academic work, I have explored how research ideas translate into real systems, developing production deployments that have allowed me to study agentic behaviour, knowledge grounding, and drift under real-world conditions. I am motivated by open questions at the intersection of agent memory, semantic alignment, and robust reasoning across languages and domains.

RESEARCH INTERESTS

Multi-Agent AI Systems • Retrieval-Augmented Generation • Large Language Models • Computational Optimization • Cyber-Physical Systems Security • AI Planning & Graph Theory

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Expected June 2026

Amrita Vishwa Vidyapeetham • Kerala, India

- Full scholarship recipient, Government of India Study in India Program
- Relevant Coursework: Machine Learning, Artificial Intelligence, Quantum Computing, Database Systems, Algorithm Design, Distributed Systems

PUBLICATIONS

Peer-Reviewed Conference Papers (Accepted)

S. Kurup et al., "When Retrieval Hurts: Evidence Utilization, Script Fidelity, and Knowledge Conflicts in Multilingual RAG," accepted for presentation at the *Workshop on Multilingual Large Language Models (MeLLMs) @ ACL 2026*.

Published

P. Rohit, A. Singh, V. Mageshen, K. Sivasabari and S. Kurup, "Hyperspectral analysis for Magnesium Estimation in Soil," 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS), Goa, India, 2024, pp. 1-4, doi: 10.1109/InGARSS61818.2024.10984012.

S. Kurup et al., "Digital Twin Framework for Precision Mushroom Cultivation Using IoT Sensors," in Proc. International Conference on Sustainable & Resilient Futures (ICSRF), 2025. (Accepted for publication)

Under Review

S. Kurup et al., "Multi-Agentic Retrieval-Augmented Generation: A Comparative Framework for Question Answering Systems," under review at CLEF 2026.

Manuscripts in Preparation

S. Kurup et al., "Advanced Control Systems for Digital Twin-Based Mushroom Cultivation: Part II," to be submitted to the International Conference on Digital Management and Information Technology, 2026.

S. Kurup et al., "Multilingual RAG Performance Analysis: Evaluation Metrics for Cross-Lingual Robustness in Low-Resource Languages," in preparation, 2026.

RESEARCH & PROFESSIONAL EXPERIENCE

Software Engineering Intern

Jan 2026 – Present

SeedlingLabs Private Limited • Bangalore, Karnataka, India

Decision Intelligence Agent: Deployed in Production (Internal)

- Designed and deployed a novel multi-agent memory layer that interconnects disparate internal applications into a unified organisational knowledge graph, with a Decision Adherence Engine that tracks, cites, and audits every strategic and operational decision company-wide
- Built a Drift Detection module that continuously monitors alignment between current actions and recorded decisions, achieving 99% detection accuracy in live production
- The system is now the company's primary institutional memory, used daily by leadership and teams; it addresses a problem that scales organisations routinely fail to solve: maintaining coherence between what was decided and what actually gets done

Agentic Educational Platform: Features Live in Institutional Production

- Co-developed a production agentic platform automating the student-teacher-parent workflow loop, including AI-generated question papers, automated answer evaluation, personalised lesson plan generation, and a real-time parent-facing performance dashboard
- Built MCP servers with CRUD guardrails to handle secure data validation and cross-application agent communication across all platform services
- The platform is live across multiple educational institutions; the primary stakeholders are teachers in under-resourced schools who lose significant teaching time to administrative work, and parents who previously had no timely visibility into their child's progress

Research Intern, Computational Cardiology

Jan 2025 – Aug 2025

Health and AI Lab, Amrita University • Kerala, India

- Applied transformer models and prompt engineering to diagnostic classification of Chagas disease from ECG signal data, and analysed epidemiological risk patterns across South American populations using neural network-based feature extraction
- Chagas disease remains severely underdiagnosed because specialist ECG interpretation is scarce in the regions most affected; the research was oriented toward making AI-assisted screening viable in low-resource clinical settings where the disease burden is highest

Summer Research Intern | Patent Pending

Jun 2025

Colorado State University • Fort Collins, Colorado, USA

- Developed an AI planning system that analyses Attack Connection Graphs and Fault Trees to identify vulnerability paths in cyber-physical systems, and built an interactive human-in-the-loop application for attack path identification and CVE-level breach detection across system layers
- Critical infrastructure operators need to reason about where their systems are most exposed before an attacker does; the tool was designed to give security teams an automated, structured way to surface and prioritise those risks rather than relying on manual review

Military Service

Oct 2020 – Oct 2022

Singapore Armed Forces • Singapore

- Completed mandatory national service, developing leadership, team coordination, and decision-making under pressure; cultivated discipline, adaptability, and stress management in demanding operational environments

TECHNICAL SKILLS

Programming & Development: Python, C++, C, TypeScript, SQL

AI/ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LlamaIndex, Scikit-learn

Agentic & Automation Systems: Multi-Agent Architectures, MCP Servers, RAG Systems, Prompt Engineering, RL, LLM Fine-tuning, Playwright, Workflow Automation

Deep Learning Architectures: 1D-CNN, ResNet, LSTM, R-LSTM, Transformer Models, Encoder-Decoder Networks, Hybrid Architectures

Quantum Computing: Qiskit, Quantum Circuits, Qubits, Quantum Gates, Grover's Algorithm

Cloud & DevOps: Docker, AWS (EC2, S3, Lambda), Google Cloud Platform, FastAPI, PostgreSQL

IoT & Hardware: MQTT, ESP32, DHT22, Sensor Integration, Real-Time Data Acquisition

Tools & Platforms: Git, FAISS, Gradio, Streamlit, Playwright, PDDL, Hyperspectral Imaging, Remote Sensing

PROJECTS

Multi-Agenting Multilingual RAG Analysis for QA Frameworks

Jan 2026 – Present

Technologies: Python, PyTorch, Hugging Face Transformers, FAISS, LangChain, ONNX Runtime

- Designed a custom metrics suite quantifying RAG model performance across 6 dimensions: semantic alignment, factual grounding, cross-lingual robustness, hallucination rate, latency, and retrieval precision
- Conducted multilingual evaluation across diverse datasets to analyse retrieval fidelity and reasoning degradation in low-resource language settings
- Most RAG benchmarks are English-centric, meaning degradation in other languages goes unmeasured; this work is motivated by making evaluation honest for the multilingual systems that researchers and practitioners actually need to deploy

Multi-Agent Retrieval-Augmented Question Answering Framework

Jul 2025 – Present

Technologies: Python, LangChain, Hugging Face Transformers, FAISS, OpenAI API, LlamaIndex, Gradio

- Built a multi-agent QA framework that separates retrieval, reasoning, and evaluation into distinct agents, with an autonomous judge and an RL-based self-improvement loop for continuous feedback optimisation
- Single-model RAG systems cannot catch their own errors because the same model that retrieves also judges its output; this architecture was motivated by the need for systems that degrade gracefully and improve from their own mistakes in production

Resiliency Graphs for Cyber-Physical Systems Preparedness

Jul 2025 – Present

Technologies: Python, NetworkX, PDDL, Graph Theory, AI Planning

- Collaborating with Colorado State University to design resiliency graphs that fuse fault trees and attack graphs, and converting Relyence fault tree data into PDDL to enable AI planning algorithms to reason over vulnerability propagation
- Infrastructure operators responsible for systems where failure has physical consequences need tools that model how risks combine and cascade, not just where individual vulnerabilities sit; this work is aimed at making that kind of structured, plannable risk analysis automatable

Digital Twin, Mushroom Cultivation | Patent Pending

Aug 2025 – Present

Technologies: Python, IoT (DHT22, ESP32), OpenCV, AWS, Matplotlib

- Developed a digital twin system that mirrors real cultivation environments using live IoT sensor data (temperature, humidity, soil moisture), with automated control of irrigation and air conditions, camera-based growth monitoring, and cloud synchronisation for remote access
- Arising from Live-in-Labs fieldwork in rural India, the system was designed so that the expertise needed to run it stays minimal; the long-term vision is enabling year-round mushroom cultivation across villages in India, turning a weather-dependent activity into a stable food and livelihood source for rural communities

Quantum Wordle Solver Using Grover's Search Algorithm

Oct 2025 – Dec 2025

Technologies: Python, Qiskit, Quantum Computing, Grover's Algorithm

- Implemented Grover's search algorithm in Qiskit with a diagonal phase oracle and diffuser circuits encoding Wordle feedback constraints into quantum states, and validated quadratic speedup via AER simulator
- The goal was to demonstrate the $O(\sqrt{N})$ speedup concretely on a constrained combinatorial problem with a verifiable classical comparison, providing an empirical reference point for quantum search applications beyond theoretical claims

AI-Driven Distributed Data-Center Scheduler

Apr 2025 – Jun 2025

Technologies: Python, PyTorch, Docker, Reinforcement Learning

- Developed a distributed simulation framework modelling multiple data centres with real-time electricity prices and renewable energy variability, and implemented an RL-based scheduler that minimises energy costs while respecting SLA and latency constraints
- Benchmarked against heuristic baselines to demonstrate improved grid flexibility; the work is relevant to energy providers and data centre operators trying to integrate renewable sources reliably without sacrificing service guarantees

LLM-Based Personalized E-Learning Platform

Feb 2025 – Jul 2025

Technologies: Python, LangChain, Streamlit, OpenAI API, PostgreSQL, FastAPI

- Built an adaptive e-learning platform that generates personalised learning pathways from user behaviour and performance analytics, with dynamic difficulty adjustment and real-time progression tracking
- Designed for learners who are left behind by one-size-fits-all curricula; the system adjusts continuously rather than waiting for a teacher to intervene, making personalisation viable at scale

Smart Agriculture Cloud Platform

Apr 2025 – Jul 2025

Technologies: Python, Google Cloud, Scikit-learn, Streamlit, MQTT

- Built a cloud-based system that processes IoT sensor data via serverless functions to generate ML-driven irrigation recommendations, with MQTT integration and a farmer-facing dashboard
- Designed for smallholder farmers who make irrigation decisions by intuition; the system gives data-grounded recommendations that can reduce water waste and improve yield reliability for those who cannot afford either

PhysioNet Challenge 2025

Jan 2025 – Aug 2025

Technologies: Python, WFDB, TensorFlow, SciPy

- Systematically compared seven deep learning architectures (1D-CNN, ResNet, LSTM, R-LSTM, LSTM-Transformer, 1D-CNN-Transformer, encoder-decoder) for Chagas disease detection from ECG signals; the hybrid LSTM-Transformer achieved the highest classification accuracy
- The aim was to identify which architecture best balances accuracy and deployability for clinical screening in low-resource settings where Chagas is most prevalent but specialist expertise is scarce

Hyperspectral Analysis for Magnesium Estimation in Soil (Published)

Jan 2024 – Aug 2024

Technologies: Python, Hyperspectral Imaging, Remote Sensing, Machine Learning, Spectral Analysis

- Developed ML models for non-invasive soil magnesium estimation using hyperspectral imaging; published at IEEE InGARSS 2024
- Conventional soil testing is slow, invasive, and expensive at scale; this work contributes to making precision agriculture accessible to farmers who cannot afford laboratory analysis by enabling field-deployable, non-destructive soil assessment

XSS Website Vulnerability Detection

Dec 2024

Technologies: Python, Scikit-learn, Random Forests, SVM

- Developed a binary classification model for detecting XSS-vulnerable websites, comparing Logistic Regression, Random Forests, and SVM with hyperparameter tuning; achieved 94% accuracy
- Manual XSS detection does not scale to the volume of sites that need assessment; this work demonstrates that systematic ML-based screening is viable as an automated first-pass security tool

LEADERSHIP & SERVICE

- Live-in-Labs Field Immersion (North India): Resided in a rural village to directly observe and understand community challenges in education and healthcare through sustained, ground-level engagement. Applied human-centred design methodologies to conduct structured needs assessments, co-ideate solutions with community members, and co-design sustainable interventions. Helped create a platform for villagers to voice concerns, surface unheard needs, and participate in shaping solutions, ensuring that proposed ideas reflected lived realities rather than outside assumptions.
- Student Social Responsibility: Designed and delivered curriculum-based STEM workshops for underprivileged students to bridge educational gaps (Aug–Nov 2024)
- Head Prefect, Amrita Vidyalayam, Kollam (2018–2020): Led student body, organised school-wide events, and mentored junior students